

Maxwell equations

Let $\mathbf{E}$ be the electric field

$$\mathbf{E} = \begin{pmatrix} A_0 \sin(kz - \omega t) \\ 0 \\ 0 \end{pmatrix}$$

and $\mathbf{B}$ the magnetic field

$$\mathbf{B} = \begin{pmatrix} 0 \\ A_0 \sin(kz - \omega t) \\ 0 \end{pmatrix}$$

where $k = \omega/c$. Verify $\mathbf{E}$ and $\mathbf{B}$ are solutions to the free-field Maxwell equations

$$\nabla \cdot \mathbf{E} = 0$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} + \frac{1}{c} \frac{\partial}{\partial t} \mathbf{B} = 0$$

$$\nabla \times \mathbf{B} - \frac{1}{c} \frac{\partial}{\partial t} \mathbf{E} = 0$$

Eigenmath script